
Beyond Error: Why Ensembles Work Until They Don't

Yash Gawankar Athena Lobo

Abstract

Ensemble methods are widely regarded as reliable performance boosters, yet their success is almost always measured by a single number: aggregate error. This scalar conflates two fundamentally different sources of generalization failure, bias and variance, making it impossible to understand why a method works or to predict when it will not. We propose studying ensemble behavior through explicit bias-variance geometry, representing each model configuration as a trajectory in the Bias²-Variance plane as key parameters are varied. Using $M=50$ bootstrap resamples, we trace these trajectories for Random Forest and XGBoost across four axes: ensemble size, base learner depth, training set size, and noise level. We then extend the analysis to hybrid prediction strategies, deriving closed-form expressions for the optimal mixing weight under both convex and probabilistic combination, and testing whether structurally different model families, including Support Vector Regression, yield better hybrids by virtue of their diversity. Experiments are conducted on synthetic regression data and validated on the California Housing dataset. Across both settings, bias-variance geometry cleanly explains not only when ensemble methods succeed but the precise conditions under which they break down, and it reveals that mixing benefit depends far less on architectural diversity than on whether parent models have genuinely complementary error profiles.

1. Introduction

Ensemble methods have become a cornerstone of modern machine learning. Bagging stabilizes unstable learners by averaging across independently trained models, while boosting constructs strong predictors by iteratively correcting the residuals of weaker ones. Methods such as Random Forest (Breiman, 2001) and XGBoost (Chen & Guestrin, 2016) consistently outperform their base learners, yet their success is evaluated almost exclusively by aggregate test error. As long as that number goes down, the question of *why* is left

unanswered.

The bias-variance decomposition provides the right language for this question (Geman et al., 1992). Under squared loss, expected prediction error decomposes into bias (systematic deviation from the true function), variance (sensitivity to the training sample), and irreducible noise. A model can achieve low error by being nearly unbiased, by being stable across training sets, or through some combination of both. Aggregate metrics make no distinction between these cases, so they cannot explain when or why a method fails under different conditions.

This paper takes a geometric view. Rather than summarizing each model as a point on a loss curve, we track how its Bias²-Variance coordinates move as key parameters change. Bagging, as realized by Random Forest, is known to reduce variance with little effect on bias (Breiman, 1996). Boosting, as realized by XGBoost, targets bias reduction through residual correction (Friedman, 2001). Our trajectory framework makes these complementary behaviors directly visible and pinpoints exactly where and why each method breaks down.

Beyond individual families, we investigate whether combining two predictors post-hoc can yield a hybrid that outperforms either parent. This question has been studied through ensemble diversity (Krogh & Vedelsby, 1995; Brown et al., 2005; Opitz & Maclin, 1999), with the intuition that uncorrelated errors cancel on averaging. We formalize this for regression by deriving the optimal mixing weight analytically under both convex and probabilistic combination strategies. In the second phase, we extend to heterogeneous pairs that include Support Vector Regression (Vapnik, 1997), testing directly whether cross-family diversity translates into mixing gain.

All experiments are evaluated on synthetic regression data and validated on the California Housing dataset (Pace & Barry, 1997), with implementation in scikit-learn (Pedregosa et al., 2011). Our contributions are: (i) bias-variance trajectory visualization as a diagnostic tool for ensemble comparison; (ii) closed-form optimal mixing weights for convex and probabilistic strategies; (iii) the finding that bias-variance *complementarity*, not model-family diversity, is the operative condition for mixing gain; and (iv) geometric evidence that aggregate error metrics structurally cannot

reveal this condition.

2. Methodology

2.1. Problem Setup and Notation

We work in supervised regression under squared loss. The expected error of predictor $\hat{f}$ decomposes as:

$$\mathbb{E}[(y - \hat{f}(x))^2] = \text{Bias}^2[\hat{f}] + \text{Var}[\hat{f}] + \sigma^2 \quad (1)$$

where $\text{Bias}[\hat{f}(x)] = \mathbb{E}[\hat{f}(x)] - f^*(x)$ is the *signed* deviation of the average prediction from the true function, and $\text{Var}[\hat{f}(x)] = \mathbb{E}[(\hat{f}(x) - \mathbb{E}[\hat{f}(x)])^2]$ measures sensitivity to the training sample. We use signed bias throughout: it distinguishes over-prediction from under-prediction, and it is essential for correctly computing the bias of a mixture. We write b_A , V_A , and V_{AB} for the signed bias, variance, and prediction covariance of model A (and B).

2.2. Empirical Estimation via Bootstrap Resampling

Each configuration is estimated using $M = 50$ independent bootstrap resamples. For each resample m , model $\hat{f}_m$ is trained and evaluated on a fixed test set. Bias and variance are estimated pointwise and averaged:

$$\widehat{\text{Bias}}(x_i) = \frac{1}{M} \sum_m \hat{f}_m(x_i) - y_i \quad (2)$$

$$\widehat{\text{Var}}(x_i) = \frac{1}{M} \sum_m (\hat{f}_m(x_i) - \bar{f}(x_i))^2 \quad (3)$$

For hybrid analysis, the *same* M samples train both models in a pair, making predictions $\hat{f}_m^A(x_i)$ and $\hat{f}_m^B(x_i)$ comparable trial-by-trial. This enables direct covariance estimation $\widehat{\text{Cov}}(A, B)$ and prediction correlation $\rho = V_{AB}/\sqrt{V_A V_B}$.

2.3. Datasets

Synthetic. We generate a 10,000-sample dataset with 15 features using scikit-learn. The target includes nonlinear terms to ensure bias is non-trivial:

$$y = \sum_i w_i x_i + 3 \sin(2x_1) + 2x_2^2 + 1.5x_3x_4 + \varepsilon \quad (4)$$

Parameters are swept one at a time: ensemble size $\in \{5, 10, 25, 50, 100, 200\}$, tree depth $\in \{2, 4, 6, 8, 10, 15, 20\}$, noise $\sigma \in \{5, 15, 30, 50, 75\}$, and training size $\in \{500, 1000, 2000, 4000, 8000\}$.

California Housing. All findings are validated on the California Housing dataset (Pace & Barry, 1997) ($\approx 20,640$ samples, 8 features), where noise is real, uncontrolled, and entangled with feature interactions.

Dataset splits. Data is partitioned before any training (Table 1). A validation set of 1,000 samples is carved before

scaling and used exclusively for hyperparameter tuning, preventing leakage. The same 4,000-sample baseline training pool is used for all models in the hybrid experiments to ensure valid covariance estimation.

Table 1. Dataset split configurations.

	Synthetic	Cal. Housing
Total samples	10,000	20,640
Test set	2,000	2,000
Validation set	1,000	1,000
Training pool	7,000	17,640
Baseline n_{train}	4,000	4,000

2.4. Models and Hyperparameter Tuning

We study Random Forest (Breiman, 2001) (bagging), XGBoost (Chen & Guestrin, 2016) (boosting), and Support Vector Regression (Vapnik, 1997) (heterogeneous mixing). We also include Bagged-XGB, which wraps XGBoost inside a BaggingRegressor. Each model is tuned on the held-out validation set *before* the bootstrap procedure begins. Tuning inside the bootstrap loop would conflate model-selection variance with prediction variance. RF is searched over depth and minimum samples per leaf; XGB over learning rate, depth, and subsample ratio (with early stopping); SVR over C , γ , and ϵ .

2.5. Convex Combination: Derivation

Given parent models A and B, the convex mixture is $\hat{f}_\lambda = \lambda \hat{f}_A + (1-\lambda) \hat{f}_B$. Since expectation is linear:

$$\text{Bias}(\hat{f}_\lambda) = \lambda b_A + (1-\lambda)b_B \quad (5)$$

$$\text{Var}(\hat{f}_\lambda) = \lambda^2 V_A + (1-\lambda)^2 V_B + 2\lambda(1-\lambda)V_{AB} \quad (6)$$

$\text{MSE}(\lambda)$ is quadratic in λ . Setting $\frac{d}{d\lambda} \text{MSE} = 0$ and solving:

$$\lambda_{\text{conv}}^* = \frac{(b_B - b_A)b_B + (V_B - V_{AB})}{(b_A - b_B)^2 + (V_A + V_B - 2V_{AB})} \quad (7)$$

The denominator equals $\text{Var}(\hat{f}_A - \hat{f}_B) \geq 0$. When the unconstrained λ^* falls outside $[0, 1]$, it is clipped to the nearest boundary, indicating no interior mixture improves on the better parent. Correlation ρ enters through $V_{AB} = \rho\sqrt{V_A V_B}$: high ρ flattens the parabola; low ρ makes an interior minimum more accessible.

2.6. Probabilistic Combination: Derivation

In the probabilistic strategy, one model is chosen per prediction via $Z \sim \text{Bernoulli}(\lambda)$. The expected prediction is identical to the convex case, so the bias is the same. The

variance differs. By the Law of Total Variance:

$$\text{Var}(\hat{f}_\lambda) = \lambda V_A + (1 - \lambda)V_B + \underbrace{\lambda(1 - \lambda)\Delta^2}_{\text{switching penalty}} \quad (8)$$

where $\Delta = b_A - b_B$ is the gap between signed biases. Expanding the total MSE and collecting terms, the cross terms $\pm 2\lambda(1 - \lambda)b_A b_B$ cancel exactly, yielding:

$$\text{MSE}_{\text{prob}}(\lambda) = \lambda \cdot \text{MSE}_A + (1 - \lambda) \cdot \text{MSE}_B \quad (9)$$

This is a key theoretical result: the probabilistic MSE is an *exact linear interpolation* of parent MSEs, regardless of bias, variance, or correlation. A linear function has no interior minimum on $[0, 1]$, so λ_{prob}^* is always a boundary solution. The probabilistic strategy can never strictly outperform the better parent.

2.7. Heterogeneous Mixing

We extend to cross-family pairs: RF+XGB, RF+SVR, and XGB+SVR. For each pair, both convex and probabilistic strategies are applied. Analytic λ_{conv}^* is computed from Eq. 7; the empirical optimum is found by grid search over 100 values in $[0, 1]$. All three models use the same 4,000-sample pool for valid covariance estimation. SVR is excluded from the training-size sweep (E4) due to $\mathcal{O}(n^2)$ memory complexity.

3. Results

3.1. Trajectory Atlas: E1–E4 (H1–H4)

E1 – Ensemble size. Figure 1 (top left) shows the clearest demonstration of the central thesis. As estimators increase from 5 to 200, Random Forest moves almost purely vertically: variance collapses from $\approx 2,850$ to ≈ 175 while Bias^2 holds steady near 4,000. XGBoost moves almost purely horizontally: Bias^2 falls from $\approx 20,000$ to ≈ 500 while variance stays between 100 and 300. The two trajectories are nearly orthogonal. At 200 estimators, XGB achieves $\text{MSE} = 753$ versus RF’s 4,253. Aggregate error reports this gap; the trajectory explains it: XGB occupies a completely different region of error space, with Bias^2 lower by a factor of eight, not a uniformly better version of the same model. Bagged-XGB clusters compactly in the lower-left throughout all four experiments.

E2 – Base learner depth. (Figure 1, top right.) RF follows a smooth diagonal sweep as depth increases: Bias^2 falls from $\approx 20,000$ to $\approx 4,000$ while variance rises gradually from ≈ 50 to ≈ 220 . XGBoost shows a non-monotone hook. At depth 2, it sits at ($\text{Bias}^2 \approx 500$, $\text{Var} \approx 180$). At depth 4, Bias^2 drops slightly and variance rises. From depth 6 onward, both Bias^2 and Variance increase together, reaching ($\text{Bias}^2 \approx 2,000$, $\text{Var} \approx 600$) at depth 20. This is not ordinary

overfitting. Boosting corrects residuals round by round: deep trees overfit those residuals, learning noise patterns rather than signal, and each subsequent round amplifies the noise, increasing both error components simultaneously. H2 is supported geometrically; the automated metric (MSE std across depths) flags “Not Supported” because RF’s large MSE range on the bias axis dominates the scalar summary.

E3 – Noise level. (Figure 1, bottom left.) RF and XGB occupy completely different regions and never approach each other. RF sits at $\text{Bias}^2 \approx 4,000$ with variance growing modestly from ≈ 215 to ≈ 315 as σ rises from 5 to 75; its bias is stable throughout. XGB sits at far lower Bias^2 but both its bias AND variance increase with noise (from ($\text{Bias}^2 \approx 380$, $\text{Var} \approx 210$) at $\sigma=5$ to ($\text{Bias}^2 \approx 600$, $\text{Var} \approx 585$) at $\sigma=75$). This dual degradation confirms H4 more strongly than predicted: boosting treats noise as signal to be corrected, iteratively reinforcing noise patterns in the mean prediction.

E4 – Training set size. (Figure 1, bottom right.) XGB’s variance drops from ≈ 835 at $n=500$ to ≈ 245 at $n=8,000$ (a $\sim 70\%$ reduction), a steep near-vertical movement in the trajectory. Its Bias^2 also drops from $\approx 2,100$ to ≈ 350 . RF moves more diagonally: Bias^2 from $\approx 9,000$ to $\approx 4,100$ and variance from ≈ 515 to ≈ 220 . The geometric pattern shows boosting is highly sensitive to training set size: sequential residual fitting depends on accurate residual estimation, which improves sharply with more data. In raw MSE terms, RF’s absolute improvement is larger (5,224 vs. 2,345) because it starts at much higher bias, not because it is more data-efficient. On California Housing, where bias levels are comparable, XGB’s MSE improvement (0.209) directly exceeds RF’s (0.193), confirming H3.

3.2. Validation on California Housing

Figure 2 shows the key trajectory patterns replicate on real data. E1 confirms the orthogonal structure: RF variance collapses from ≈ 0.047 to ≈ 0.005 as estimators grow, while XGB Bias^2 falls from ≈ 0.78 to ≈ 0.22 . E2 reproduces the depth instability, with XGB variance rising sharply from ≈ 0.005 at depth 2 to ≈ 0.066 at depth 20 while its Bias^2 stays nearly flat past depth 6. The Cal Housing pattern is therefore a variance explosion rather than the bidirectional U-turn seen on synthetic, but it is the same fragility: XGB is far more depth-sensitive than RF. One notable difference between datasets: RF and XGB are much closer in MSE on real data (RF = 0.261, XGB = 0.238, factor of 1.1) compared to synthetic (factor of 5.6). This compression has direct consequences for the mixing analysis.

E4 (Figure 2, right) confirms H3 on real data: XGB benefits along both error axes simultaneously ($\text{Bias}^2: \approx 0.46 \rightarrow 0.25$; Variance: $\approx 0.076 \rightarrow 0.021$), while RF — already low-variance by design — gains only in bias reduction. XGB’s

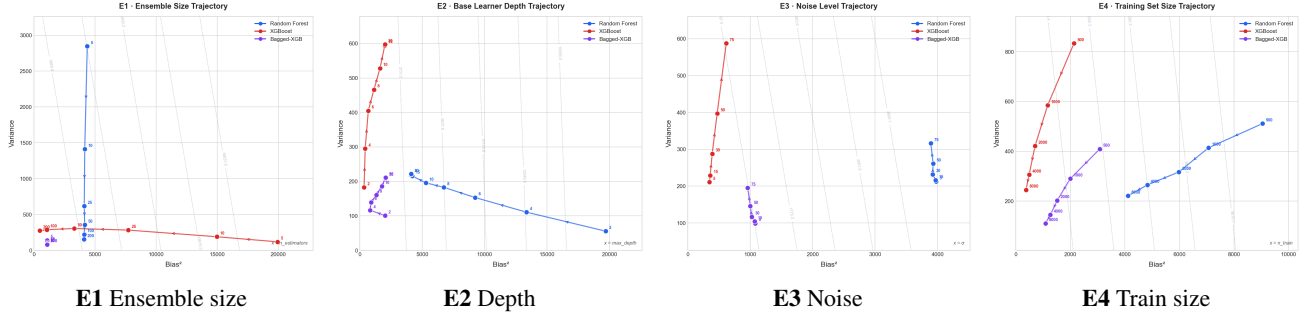


Figure 1. Trajectory atlas on synthetic data (Bias² vs. Variance plane). **E1:** RF moves vertically (variance only); XGB horizontally (bias only) — orthogonal mechanisms. **E2:** XGB U-turns from depth 6: both components increase simultaneously. **E3:** XGB degrades in both components under noise; RF bias is stable. **E4:** XGB descends diagonally with more data; RF gains only in bias. Bagged-XGB (purple) stays lower-left across all experiments.

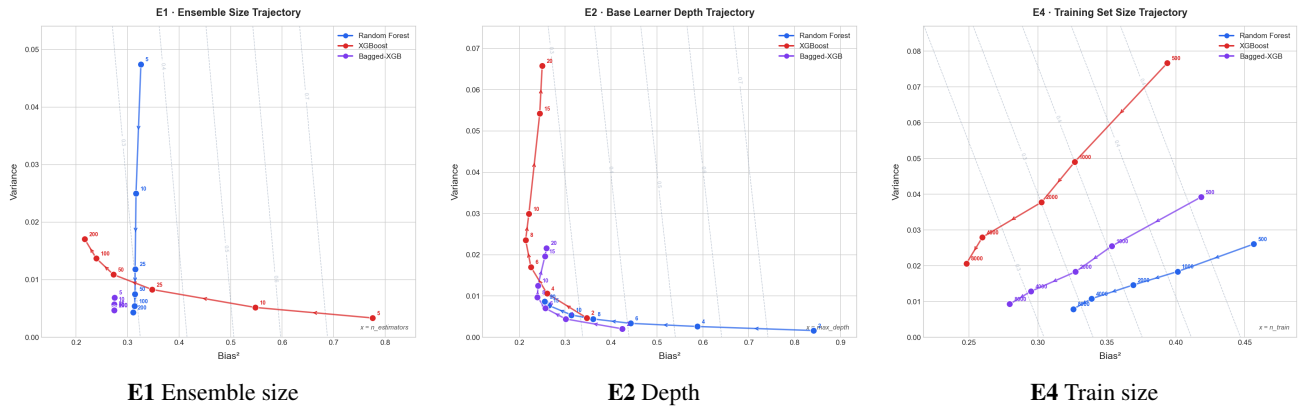


Figure 2. California Housing trajectory validation (E1, E2, E4). Orthogonal mechanisms and depth sensitivity replicate on real data. E4: XGB descends diagonally (both Bias² and Variance shrink); RF moves nearly horizontally, confirming H3. RF and XGB are much closer in absolute MSE than on synthetic data (factor 1.1 vs. 5.6), which drives the mixing results.

MSE improves by 0.209 vs. RF's 0.193 from $n=500$ to $n=8,000$.

3.3. Homogeneous Mixing (H5–H7)

Figure 3 shows the mixing analysis for RF+XGB. On both synthetic and California Housing data, the MSE curve descends monotonically from pure RF to pure XGB. Convex and probabilistic strategies produce visually indistinguishable curves and both clip to $\lambda^*=1.0$ (pure XGB). No mixture beats either parent on either dataset. The synthetic theory formula gives $\lambda^*_{\text{conv}}=0.31$, but the unconstrained parabola vertex at that point still lies far above XGB's boundary MSE. When one parent is overwhelmingly better, the formula is mathematically correct but empirically irrelevant. H5 (mixture beats both parents) and H7 (convex beats probabilistic) are refuted on both datasets. H6 (theory λ^* matches empirical) is also refuted: the synthetic gap of 0.69 reflects the extreme flatness of the MSE parabola, where the closed-form formula becomes numerically unreliable.

3.4. Heterogeneous Mixing: Diversity Without Complementarity (H8–H9)

Figure 4 presents the most counterintuitive findings of the paper. The left panel (Plot 7) shows prediction correlations: $\rho(\text{RF}, \text{XGB}) = 0.248$ on synthetic (0.529 on California Housing), while both SVR pairs have $\rho \approx 0$ on both datasets. The SVR pairs are statistically independent of the tree models, which is precisely what the diversity hypothesis predicts should produce better hybrids.

The right panel (Plot 8, California Housing) shows the outcome. RF+XGB (green) descends monotonically to pure XGB with no meaningful mixing improvement, consistent with the homogeneous analysis. Both SVR pairs show the opposite: MSE increases steeply as SVR weight grows. On California Housing, SVR achieves MSE = 0.387, worse than both tree models. On synthetic data, SVR achieves MSE = 234, better than both (all curves point toward SVR as the better parent). The correlation structure is identical in both cases, yet the direction of optimal mixing flips en-

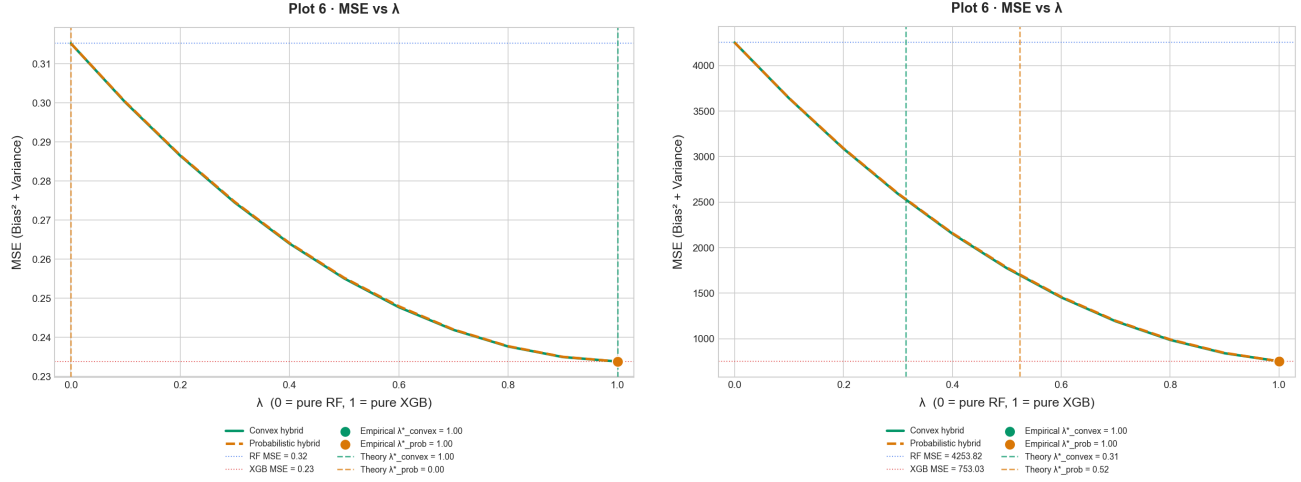


Figure 3. MSE vs. λ for RF+XGB. **Left (California Housing):** MSE descends smoothly from ≈ 0.315 (pure RF) to ≈ 0.234 (pure XGB); the empirical and theoretical optimum for both strategies is $\lambda^* = 1.0$. **Right (Synthetic):** MSE descends from 4,253 (pure RF) to 753 (pure XGB); both strategies again clip to $\lambda^* = 1.0$. In both datasets the convex and probabilistic curves are visually indistinguishable, and the boundary solution (pure XGB) is optimal.

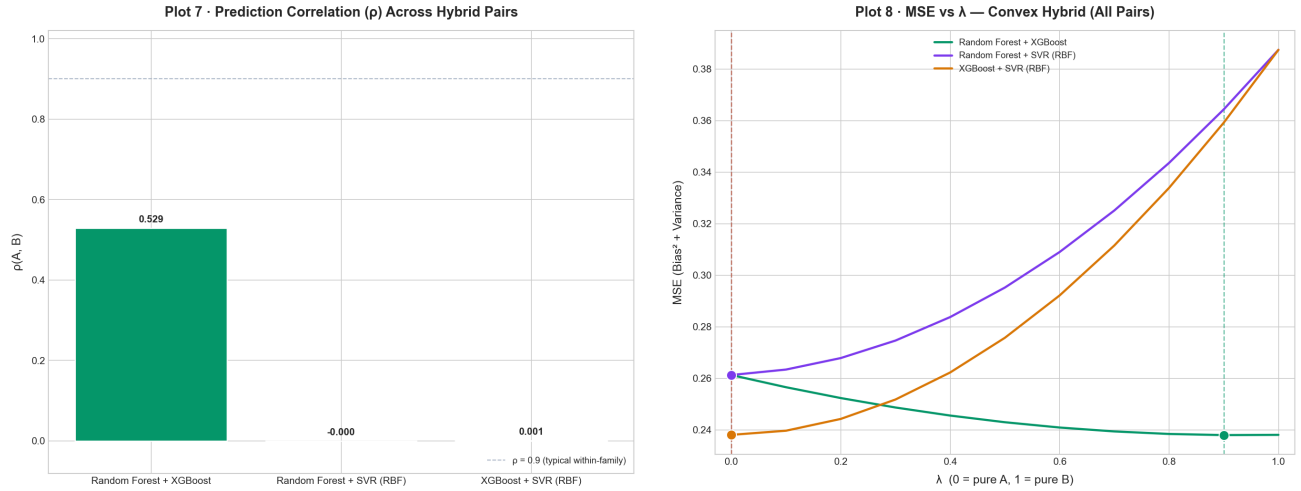


Figure 4. **Left:** Prediction correlation across pairs on California Housing. RF+XGB has $\rho=0.529$; both SVR pairs have $\rho \approx 0$, indicating statistical independence from the tree models. The dashed reference line at $\rho=0.9$ marks a typical literature benchmark for within-family ensembles. **Right (California Housing):** MSE vs. λ for all pairs. RF+XGB (green) descends to pure XGB with no meaningful mixing improvement. Both SVR pairs increase steeply away from the tree models. Lower ρ does not predict mixing gain.

tirely. H8 is not supported: lower ρ did not predict greater mixing reduction. The homogeneous RF+XGB pair, with the *highest* correlation, was the only pair to show any mixing improvement. H9 is also not supported; the convex advantage over probabilistic was zero across all pairs.

The unifying explanation is complementarity, not diversity. Low ρ widens the variance-reduction channel in the convex formula (reducing V_{AB}) and flattens the MSE parabola. But this only produces an interior minimum when parent MSEs are comparable *and* error profiles are offsetting. When one parent dominates, λ^* is driven to a boundary by the bias

gap, and no covariance structure can help. The geometry of Eq. 7 makes this explicit.

3.5. Hypothesis Verdict Summary

4. Process

Framework design. The trajectory view emerged from the question of whether bagging and boosting’s complementary mechanisms could be made directly visible. Shared $M=50$ bootstrap samples across model pairs were essential for valid covariance estimation. Early runs on linear targets

Table 2. Hypothesis verdicts across both datasets.

Hyp.	Description	Syn.	Cal.H.
H1	Orthogonal trajectories (Bias ² -Var)	Conf.	Conf.
H2	Boosting fragile to depth (U-turn)	Conf. [†]	Conf. [†]
H3	Boosting benefits more from data	Partial [‡]	Conf.
H4	Noise degrades boosting in both	Conf.	N/A
H5	Mixture beats both parents	Refuted	Refuted
H6	Theory λ^* matches empirical	Refuted	Refuted
H7	Convex beats probabilistic	Refuted	Refuted
H8	Cross-family diversity $\rightarrow$ gain	Refuted	Refuted
H9	Convex adv. amplified by D^2	Refuted	Refuted

[†]Geometric verdict; MSE-std metric misleading (axis-scale asymmetry).

[‡]XGB variance drops steeply (n=500: Var $\approx$ 835 $\rightarrow$ n=8000: $\approx$ 245, $\sim$ 70% reduction); RF MSE improvement larger only because of high initial bias.

produced compressed trajectories; the nonlinear target function was added to give the bias axis meaningful spread. Two key implementation choices proved consequential: the validation set is carved before scaling and used only for tuning (tuning inside the bootstrap loop conflates model-selection variance with prediction variance), and signed bias is used throughout (unsigned bias breaks hybrid bias computation since cancellation effects are lost when $\lambda b_A + (1-\lambda)b_B$ is computed from absolute values).

Surprises in the results. H2 predicted a variance explosion under deep XGB trees. The plot showed something different: from depth 6 onward, both Bias² and Variance increase simultaneously. Deep trees overfit individual residuals; subsequent boosting rounds amplify noise rather than correct signal, shifting the mean prediction away from the true function. This U-turn emerged from the geometry, not prior theory. For heterogeneous mixing, the initial expectation was that lower ρ would predict better hybrids. The results refuted this cleanly: the RF+XGB pair with $\rho=0.529$ was the only pair to show any mixing improvement, while both SVR pairs with $\rho\approx 0$ showed none. The complementarity condition - comparable parent MSEs and offsetting Bias²/Variance profiles emerged post-hoc: we asked why low ρ failed to predict mixing gain, and found that the SVR pairs violated both conditions (SVR either dominated or was far worse), making the correlation structure irrelevant.

SVR tuning and the directional flip. With default $C=1.0$, SVR produced MSE = 12,490 on the synthetic dataset (worse than RF). Validation-set tuning identified $C=1,000$, yielding MSE = 234 (better than both tree models). On

California Housing, SVR achieves MSE = 0.387 (worse than both). This performance reversal across datasets, while the correlation structure ($\rho \approx 0$ for all SVR pairs) stays identical, is what makes the comparison informative: diversity alone is not sufficient.

5. Contributions

We built a complete bias-variance decomposition pipeline from scratch in Python. The core is a bootstrap resampling evaluator with shared sampling across model pairs for covariance estimation, supporting parameter sweeps across all four axes. The hybrid module implements convex and probabilistic strategies with closed-form λ^* and grid-search empirical validation. Tuning is a one-shot validation-set search before bootstrap. No starter code was used. All modelling uses scikit-learn (Pedregosa et al., 2011) and xgboost (Chen & Guestrin, 2016); computation uses NumPy/SciPy; visualization uses Matplotlib. The full pipeline is available at: <https://github.com/yashgawankar/Ensembles>.

6. AI Usage

We used Claude (Anthropic) for code infrastructure: boilerplate for result storage, argument parsing, and plot scaffolding. All generated code was reviewed and corrected; one fix involved an incorrect seed-matching scheme in the SVR integration that would have invalidated covariance estimates. AI also assisted with documentation and the LaTeX template structure.

The research questions, all nine hypotheses, and mathematical derivations (convex λ^* and the probabilistic linearity proof) were developed independently. All result interpretation was ours; the complementarity finding emerged from examining plots directly. AI helped identify classical references, which were manually verified before inclusion. We treated all AI output as a starting draft requiring critical review.

References

- Breiman, L. Bagging predictors. *Machine Learning*, 24(2): 123–140, 1996.
- Breiman, L. Random forests. *Machine Learning*, 45(1): 5–32, 2001.
- Brown, G., Wyatt, J., Harris, R., and Yao, X. Diversity creation methods: A survey and categorisation. *Information Fusion*, 6(1):5–20, 2005.
- Chen, T. and Guestrin, C. XGBoost: A scalable tree boosting system. In *Proceedings of the 22nd ACM SIGKDD*

International Conference on Knowledge Discovery and Data Mining, pp. 785–794, 2016.

Friedman, J. H. Greedy function approximation: A gradient boosting machine. *Annals of Statistics*, 29(5):1189–1232, 2001.

Geman, S., Bienenstock, E., and Doursat, R. Neural networks and the bias/variance dilemma. *Neural Computation*, 4(1):1–58, 1992.

Krogh, A. and Vedelsby, J. Neural network ensembles, cross validation, and active learning. In *Advances in Neural Information Processing Systems*, volume 7, 1995.

Opitz, D. and Maclin, R. Popular ensemble methods: An empirical study. *Journal of Artificial Intelligence Research*, 11:169–198, 1999.

Pace, R. K. and Barry, R. Sparse spatial autoregressions. *Statistics & Probability Letters*, 33(3):291–297, 1997.

Pedregosa, F. et al. Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research*, 12:2825–2830, 2011.

Vapnik, V. The support vector method. In *International Conference on Artificial Neural Networks*, pp. 261–271. Springer, 1997.